

Notes: Antràs and Chor (Econometrica, 2013)

Organizing the Global Value Chain

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1 Big picture

- **Question.** How should a firm allocate ownership rights along a *sequential* production chain when suppliers must make relationship-specific investments and contracts are incomplete?
- **Why this is new.** Standard property-rights models typically study a single supplier or a set of symmetric suppliers. This paper adds a technologically ordered continuum of stages, so the same input may be more or less upstream or downstream, and that position itself affects make-or-buy decisions.
- **Core friction.** A final-good producer cannot write enforceable contracts on whether a supplier's input is actually compatible with the rest of the chain. Ex post bargaining therefore determines how the marginal surplus from each stage is split.
- **Trade-off.** Integration raises the firm's bargaining share and helps it extract rents, but it lowers the supplier's incentive to invest. Outsourcing does the opposite.
- **Main theoretical result.** The optimal ownership pattern depends on the stage's *downstreamness* and on the comparison between demand curvature and technological substitutability:
 - if demand is sufficiently elastic relative to input substitutability (the paper's *sequential complements* case, $\rho > \alpha$), the firm outsources relatively *upstream* stages and integrates relatively *downstream* stages;
 - if demand is sufficiently inelastic relative to input substitutability (the *sequential substitutes* case, $\rho < \alpha$), the pattern reverses: the firm integrates *upstream* stages and outsources *downstream* stages.
- **Key intuition.** Upstream investment affects all later stages. When upstream investment makes later investment more attractive, the firm wants to protect upstream incentives and postpone integration toward the end of the chain. When upstream investment instead depresses the marginal return of later investment, the firm wants more control early on.
- **Empirical idea.** Use U.S. related-party import shares as an industry-level proxy for the prevalence of integration, construct industries' average position in the value chain from the 2002 U.S. Input–Output Tables, and interact downstreamness with a proxy for the average buyer demand elasticity based on Broda and Weinstein (2006).
- **Main empirical result.** In industries whose buyers face high demand elasticities, more downstream inputs are imported more within firm boundaries. In industries whose buyers face low demand elasticities, the sign flips or becomes weakly negative. The strongest results appear in weighted regressions that downweight very small trade flows.
- **Economic magnitude.** In the weighted baseline, a one-standard-deviation increase in `DUse_TUse` raises the intrafirm import share by about $0.482 \times 0.228 = 0.110$ in the high-elasticity group. Using `DownMeasure`, the analogous effect is about $0.526 \times 0.222 = 0.117$.
- **Broader message.** The boundary of the multinational firm is not only about industry-level headquarter intensity or contractibility. It also depends on *where* an input sits in the production line.

2 Data and measurement (key objects)

2.1 Main trade data

- **Source.** U.S. Census Bureau Related Party Trade Database, 2000–2010.
- **Coverage used in the paper.** Manufacturing industries mapped to 2002 NAICS-based IO industries. The working sample contains **253 manufacturing industries** with intrafirm import information.
- **Unit of observation.**
 - baseline: industry-year,
 - richer specification: source-country $\times$ industry $\times$ year.
- **Dependent variable.**

$$S_{it} = \frac{\text{U.S. related-party imports of industry } i \text{ in year } t}{\text{total U.S. imports of industry } i \text{ in year } t}. \quad (1)$$

At the country level, the analogous object is S_{ict} .

- **Measurement detail.** A tiny “unreported” category exists when the related-party status of a flow is not known. It is less than 0.2% of total U.S. manufacturing imports, but it matters more for small trade flows, which is why weighted regressions are important later.

2.2 Downstreamness: the paper’s key explanatory variable

The paper develops two industry-level measures of position in the value chain using the 2002 U.S. Input–Output Tables.

Let Y_i be gross output of industry i , F_i the portion going to final demand, and let D be the matrix of direct input requirements. The basic identity is

$$Y = F + DF + D^2F + D^3F + \dots = [I - D]^{-1}F. \quad (30)$$

- **Measure 1: DUse_TUse.** This is the ratio of aggregate *direct use* of an industry’s output in final-use production to its aggregate *total use* as an input.

$$\text{DUse_TUse}_i \equiv \frac{[DF]_i}{[Y - F]_i}. \quad (2)$$

Interpretation. A high value means the industry’s output tends to be used close to final demand, so it is relatively downstream.

- **Measure 2: DownMeasure.** The paper first defines an upstreamness measure using the weighted sum

$$F + 2DF + 3D^2F + 4D^3F + \dots = [I - D]^{-2}F. \quad (31)$$

For industry i , upstreamness is

$$U_i = \frac{([I - D]^{-2}F)_i}{Y_i}, \quad (3)$$

and the downstreamness measure used in the regressions is its reciprocal:

$$\text{DownMeasure}_i = \frac{1}{U_i}. \quad (4)$$

Interpretation. A higher value again means the industry’s output is used closer to final demand on average.

2.3 Demand-elasticity proxy used to separate theory cases

- The theory distinguishes the cases $\rho > \alpha$ and $\rho < \alpha$, where the price elasticity of demand is $1/(1 - \rho)$.
- The data do *not* contain good direct estimates of cross-input substitutability α by industry.
- The empirical workaround is to compute, for each seller industry i , a weighted average of the import-demand elasticities of the industries that buy i ’s output, using Broda and Weinstein (2006) elasticities mapped up from HS10 products.
- The paper then splits industries by the median or by quintiles of this average buyer demand elasticity proxy.
- **Interpretation.** High average buyer demand elasticity is treated as the empirically relevant version of the complements case; low average buyer demand elasticity is treated as the substitutes case.

2.4 Controls that matter

The main control vector includes *average buyer industry* characteristics rather than seller-industry characteristics.

- skill intensity: $\log(s/\ell)$,
- equipment capital intensity: $\log(\text{equipment } k/\ell)$,
- plant capital intensity: $\log(\text{plant } k/\ell)$,
- materials intensity,
- R&D intensity: $\log(0.001 + \text{R\&D}/\text{Sales})$,
- firm-size dispersion.

These come directly from the model extensions: headquarter intensity should increase integration, and productivity dispersion should raise the prevalence of integration when fixed costs make only the most profitable firms integrate.

2.5 Objects that are observed versus unobserved

- **Observed.** Related-party import shares, source countries, industries’ input-output positions, buyer-side factor intensities, buyer-side demand elasticity proxies, intermediation and contractibility variables, and country entry-cost variables used in the selection correction.

- **Unobserved.** Stage-level ownership choices within a specific firm, exact technological substitutability α , exact stage ordering within each firm's own chain, and firm-specific productivity thresholds for integration.
- **Implication.** The empirics necessarily aggregate a firm-level theory to the industry level. The paper is very explicit about this limitation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Benchmark model: sequential production

A final-good firm produces using a continuum of stages indexed by $j \in [0, 1]$, with larger j meaning more downstream stages.

$$q = \theta \left(\int_0^1 x(j)^\alpha I(j) dj \right)^{1/\alpha}, \quad \alpha \in (0, 1). \quad (1)$$

Here $x(j)$ is the service flow of a compatible stage- j input, and the indicator $I(j)$ equals 1 if stage j is completed only after all upstream stages have been completed, and 0 otherwise.

Interpretation.

- Production is technologically sequential.
- Inputs are physically complementary in the engineering sense, but they enter a CES aggregator, so the quality-adjusted value of output still allows some substitution across stage qualities.
- Stage position matters because downstream production cannot begin before upstream delivery.

3.2 Demand and revenue

Consumers have CES-like demand over final-good varieties:

$$U = \left(\int_{\omega \in \Omega} (\varphi(\omega) \tilde{q}(\omega))^\rho d\omega \right)^{1/\rho}, \quad \rho \in (0, 1). \quad (2)$$

This implies a constant price elasticity of demand equal to

$$\varepsilon = \frac{1}{1 - \rho}. \quad (5)$$

Combining this with the production function yields firm revenue

$$r = A^{1-\rho} \theta^\rho \left(\int_0^1 x(j)^\alpha I(j) dj \right)^{\rho/\alpha}, \quad (3)$$

where $A > 0$ is an industry demand shifter.

Interpretation.

- α governs technological substitutability across stage inputs.
- ρ governs demand curvature.
- The key theoretical comparison is ρ versus α .

3.3 Complete-contract benchmark

With complete contracts, the firm can dictate all stage investments:

$$\max_{\{x(j), s(j)\}} A^{1-\rho} \theta^\rho \left(\int_0^1 x(j)^\alpha dj \right)^{\rho/\alpha} - \int_0^1 s(j) dj \quad \text{s.t.} \quad s(j) - cx(j) \geq 0. \quad (6)$$

The solution equalizes investment across stages:

$$x^C = \left(\frac{\rho A^{1-\rho} \theta^\rho}{c} \right)^{1/(1-\rho)}. \quad (7)$$

Associated profits are

$$\pi^C = (1 - \rho) A \left(\frac{\rho \theta^\rho}{c^\rho} \right)^{1/(1-\rho)}. \quad (8)$$

Interpretation.

- Under full contractibility, there is no reason to vary ownership or incentives across stages.
- All the interesting ownership patterns later are purely due to incomplete contracting.

3.4 Incomplete contracts, ex post bargaining, and supplier incentives

Contracts cannot be written on whether the delivered input is actually compatible with the chain. The firm can only specify whether the supplier is integrated or independent.

Let revenue secured up to stage m be

$$r(m) = A^{1-\rho} \theta^\rho \left(\int_0^m x(j)^\alpha dj \right)^{\rho/\alpha}. \quad (4)$$

The stage- m supplier's marginal contribution is

$$r'(m) = \frac{\rho}{\alpha} \left(A^{1-\rho} \theta^\rho \right)^{\alpha/\rho} r(m)^{(\rho-\alpha)/\rho} x(m)^\alpha. \quad (5)$$

Ownership changes the firm's bargaining share:

$$\beta(m) = \begin{cases} \beta_O, & \text{if stage } m \text{ is outsourced,} \\ \beta_V > \beta_O, & \text{if stage } m \text{ is integrated.} \end{cases} \quad (9)$$

The supplier gets the remaining share $1 - \beta(m)$ and solves

$$\max_{x(m)} \pi_S(m) = (1 - \beta(m)) \frac{\rho}{\alpha} \left(A^{1-\rho} \theta^\rho \right)^{\alpha/\rho} r(m)^{(\rho-\alpha)/\rho} x(m)^\alpha - cx(m). \quad (6)$$

The first-order condition gives

$$x(m) = \left[(1 - \beta(m)) \frac{\rho (A^{1-\rho} \theta \rho)^{\alpha/\rho}}{c} \right]^{1/(1-\alpha)} r(m)^{(\rho-\alpha)/(\rho(1-\alpha))}. \quad (7)$$

Interpretation.

- Outsourcing raises the supplier's bargaining share and therefore raises $x(m)$.
- Integration does the reverse: it weakens the supplier's incentive to invest, but strengthens the firm's rent-extraction motive.
- The novel object is the dependence of $x(m)$ on $r(m)$, which is cumulative value generated up to stage m .

3.5 Sequential complements versus sequential substitutes

Plugging (7) into (5) yields

$$r'(m) = \frac{\rho}{\alpha} \left(\frac{(1 - \beta(m)) \rho \theta}{c} \right)^{\alpha/(1-\alpha)} A^{\alpha(1-\rho)/(\rho(1-\alpha))} r(m)^{(\rho-\alpha)/(\rho(1-\alpha))}. \quad (8)$$

Solving this differential equation gives

$$r(m) = A \left(\frac{1 - \rho}{1 - \alpha} \right)^{\rho(1-\alpha)/(\alpha(1-\rho))} \left(\frac{\rho \theta}{c} \right)^{\rho/(1-\rho)} \left[\int_0^m (1 - \beta(j))^{\alpha/(1-\alpha)} dj \right]^{\rho(1-\alpha)/(\alpha(1-\rho))}. \quad (9)$$

Hence equilibrium investment along the chain is

$$\begin{aligned} x(m) = & A \left(\frac{1 - \rho}{1 - \alpha} \right)^{(\rho-\alpha)/(\alpha(1-\rho))} \left(\frac{\rho}{c} \right)^{1/(1-\rho)} \theta^{\rho/(1-\rho)} (1 - \beta(m))^{1/(1-\alpha)} \\ & \times \left[\int_0^m (1 - \beta(j))^{\alpha/(1-\alpha)} dj \right]^{(\rho-\alpha)/(\alpha(1-\rho))}. \end{aligned} \quad (10)$$

Key interpretation.

- If $\rho > \alpha$, the exponent on the integral term is positive. More upstream investment raises the return to downstream investment: **sequential complements**.
- If $\rho < \alpha$, that exponent is negative. More upstream investment lowers the return to downstream investment: **sequential substitutes**.
- This is the paper's main conceptual twist. Inputs are technologically complementary, but when demand is sufficiently concave, revenue curvature can make investments behave like substitutes in quality-adjusted value terms.

3.6 Firm profit and the optimal organizational structure

Firm profit net of supplier payments is

$$\begin{aligned} \pi_F = & A \frac{\rho}{\alpha} \left(\frac{1 - \rho}{1 - \alpha} \right)^{(\rho-\alpha)/(\alpha(1-\rho))} \left(\frac{\rho \theta}{c} \right)^{\rho/(1-\rho)} \\ & \times \int_0^1 \beta(j) (1 - \beta(j))^{\alpha/(1-\alpha)} \left[\int_0^j (1 - \beta(k))^{\alpha/(1-\alpha)} dk \right]^{(\rho-\alpha)/(\alpha(1-\rho))} dj, \end{aligned} \quad (11)$$

subject to $\beta(j) \in \{\beta_V, \beta_O\}$.

To study the relaxed problem, define

$$v(j) = \int_0^j (1 - \beta(k))^{\alpha/(1-\alpha)} dk. \quad (12)$$

The firm then solves

$$\pi_F(v) = \kappa \int_0^1 (1 - v'(j))^{(1-\alpha)/\alpha} v'(j) v(j)^{(\rho-\alpha)/(\alpha(1-\rho))} dj, \quad (13)$$

for a positive constant κ , yielding the Euler–Lagrange condition

$$v^{(\rho-\alpha)/(\alpha(1-\rho))} (v')^{(1-\alpha)/\alpha-1} \left[v'' + \frac{\rho - \alpha}{1 - \rho} \frac{(v')^2}{v} \right] = 0. \quad (14)$$

Imposing the boundary conditions, the unconstrained optimal division of surplus is

$$\beta^*(m) = 1 - \alpha m^{(\alpha-\rho)/\alpha}. \quad (15)$$

This is the sharpest theoretical equation in the paper.

- If $\rho > \alpha$, then $\beta^*(m)$ rises with m : the firm wants more bargaining power later in the chain.
- If $\rho < \alpha$, then $\beta^*(m)$ falls with m : the firm wants more bargaining power earlier in the chain.

This yields the paper’s main propositions.

- **Proposition 1.** $\beta^*(m)$ is increasing in m in the complements case and decreasing in m in the substitutes case.
- **Proposition 2.** In the complements case, there is a cutoff m_c^* such that upstream stages are outsourced and downstream stages are integrated. In the substitutes case, there is a cutoff m_s^* such that upstream stages are integrated and downstream stages are outsourced.
- **Proposition 3.** Whenever integration and outsourcing coexist, a fall in ρ expands the range of integrated stages. Lower demand elasticity means more market power, so the firm puts more weight on rent extraction.

The paper derives closed-form formulas for m_c^* and m_s^* (its equations (16)–(17)). The economic content is more important than the algebra: the integrated set is a connected interval, and whether it lies upstream or downstream is governed entirely by the sign of $\rho - \alpha$.

3.7 Important extensions

3.7.1 Ex ante transfers

- If ex ante lump-sum transfers are allowed, ownership no longer serves rent extraction directly because surplus can be redistributed up front.
- The qualitative ranking of integration incentives still follows downstreamness, but coexistence of integration and outsourcing survives only in the substitutes case.
- **Interpretation.** This extension is useful because it shows exactly which parts of the benchmark are driven by bargaining-rent extraction and which parts are driven by investment-incentive management.

3.7.2 Linkages across bargaining rounds

- The benchmark assumes each stage bargains only over its own marginal contribution.
- The paper solves a richer sequential bargaining game in which a supplier may internalize consequences for later bargaining rounds.
- In the continuum limit, the equilibrium investment choices coincide with the benchmark. So the main results are not an artifact of the stage-by-stage bargaining simplification.

3.7.3 Headquarter intensity

The production function becomes

$$q = \theta \left(\frac{h}{\eta} \right)^\eta \left(\int_0^1 \left(\frac{x(j)}{1-\eta} \right)^\alpha I(j) dj \right)^{(1-\eta)/\alpha}, \quad \eta \in (0, 1). \quad (22)$$

- Headquarter services h are chosen by the firm before suppliers invest.
- The effective revenue-curvature parameter becomes $\tilde{\rho} = (1 - \eta)\rho$.
- **Proposition 4.** The complements/substitutes comparison becomes $\tilde{\rho} > \alpha$ versus $\tilde{\rho} < \alpha$, and higher headquarter intensity η also shifts the level of the integration decision upward.
- **Interpretation.** Headquarter-intensive industries should integrate more, but high headquarter intensity also makes it harder for downstreamness to have a positive slope, because it lowers effective ρ .

3.7.4 Firm heterogeneity and the prevalence of integration

Productivity is Pareto distributed:

$$G(\theta) = 1 - (\underline{\theta}/\theta)^z, \quad \theta \geq \underline{\theta} > 0. \quad (23)$$

Integrated relationships also entail higher fixed costs, with $f_V > f_O$.

- In the complements case, more productive firms integrate a larger downstream interval.
- In the substitutes case, more productive firms integrate a larger upstream interval.

If $\theta_c(m)$ is the productivity cutoff for integrating stage m in the complements case, then the share of firms integrating stage m is

$$\sigma_c(m) = \begin{cases} 0, & m < m_c^*, \\ (\underline{\theta}/\theta_c(m))^z, & m > m_c^*. \end{cases} \quad (25)$$

Proposition 5.

- The *industry-level prevalence* of integration is increasing in downstreamness when $\rho > \alpha$.
- It is decreasing in downstreamness when $\rho < \alpha$.

- It is also increasing in within-industry productivity dispersion.

Why this matters. This is the bridge from the firm's stage-level theory to the industry's intrafirm import share in the data.

3.7.5 Input and supplier heterogeneity

Allow stage-specific productivity and costs:

$$q = \theta \left(\int_0^1 (\psi(j)x(j))^\alpha I(j) dj \right)^{1/\alpha}, \quad (26)$$

with marginal cost $c(j)$.

Then the optimal-surplus-share derivative satisfies

$$\frac{d\beta^*(m)}{dm} = \frac{(\rho - \alpha)(1 - \alpha)}{(1 - \rho)\alpha} (1 - \beta(m)) \frac{\left(\frac{1 - \beta(m)}{c(m)/\psi(m)} \right)^{\alpha/(1-\alpha)}}{\int_0^m \left(\frac{1 - \beta(j)}{c(j)/\psi(j)} \right)^{\alpha/(1-\alpha)} dj}. \quad (28)$$

Proposition 6. Despite heterogeneity in stage importance and stage cost, the sign of the effect of downstreamness on integration is still governed by $\rho - \alpha$.

3.8 Empirical implementation of the theory

The baseline regression is

$$S_{it} = \beta_1 D_i \mathbf{1}(\text{Elas}_i < \text{Median}) + \beta_2 D_i \mathbf{1}(\text{Elas}_i > \text{Median}) + \delta \mathbf{1}(\text{Elas}_i > \text{Median}) + X'_{it} \Gamma + \alpha_t + \varepsilon_{it}. \quad (32)$$

The richer country-industry-year version is

$$S_{ict} = \beta_1 D_i \mathbf{1}(\text{Elas}_i < \text{Median}) + \beta_2 D_i \mathbf{1}(\text{Elas}_i > \text{Median}) + \delta \mathbf{1}(\text{Elas}_i > \text{Median}) + X'_{it} \Gamma + \alpha_{ct} + \varepsilon_{ict}. \quad (33)$$

The more flexible quintile specification is

$$S_{it} = \sum_{s=1}^5 \gamma_s D_i \mathbf{1}(\text{Elas}_i \in Q_s) + \sum_{s=2}^5 \delta_s \mathbf{1}(\text{Elas}_i \in Q_s) + X'_{it} \Gamma + \alpha_t + \varepsilon_{it}. \quad (34)$$

Interpretation of the coefficients.

- The theory predicts $\beta_1 < 0$ and $\beta_2 > 0$.
- The weighted regressions are not a new structural model; they are a measurement-error correction device because related-party status is noisier for small trade flows.
- Country-year fixed effects absorb source-country shocks common to all industries: factor prices, exchange rates, country institutions, and macro conditions.

4 Identification and what makes the estimates credible

- **The paper is identified off a sign reversal, not off a single slope.** The cleanest prediction is not “downstreamness increases integration” in general; it is that the slope depends on the demand-elasticity environment. That is a stronger and more falsifiable comparative-static test.
- **Downstreamness is measured independently from intrafirm trade.** The key regressors come from the 2002 U.S. Input–Output Tables, whereas the dependent variable comes from U.S. related-party imports over 2000–2010.
- **Two different downstreamness measures deliver similar messages.** `DUse_TUse` captures closeness to final use in a direct-use sense; `DownMeasure` uses the entire distribution of distances to final demand. Agreement across the two makes the story more credible.
- **Buyer-side controls fit the theory better than seller-side controls.** The sourcing decision belongs to the final-good producer. Using average buyer factor intensities, R&D, and dispersion makes the controls structurally aligned with the model.
- **Country-year fixed effects are powerful.** In the country-level regressions, these fixed effects soak up many alternative explanations based on source-country conditions.
- **Weighted regressions address a real data problem.** The related-party status of very small flows is measured less cleanly because of the “unreported” category. Weighting by trade value makes the core results stronger, exactly as one would expect if the main issue were classical measurement error.
- **Robustness is broad.** The results survive controls for value-added shares, input importance, intermediation, own and buyer contractibility, and alternative sample splits by elasticity quintiles.
- **Selection into importing is addressed directly.** The paper runs a Heckman correction at the country-industry-year level, using an interaction of country entry costs with seller-industry R&D intensity in the selection equation.
- **Main limitation.** The model is at the firm–stage level, but the evidence is industry-level. The authors are upfront that better firm-level data on input-level sourcing would be ideal.
- **Another limitation.** The empirical split between complements and substitutes is based on buyer demand elasticity rather than direct estimates of cross-input substitutability. That is sensible, but imperfect.

5 Tables and figures

5.1 Figure 1: descriptive sign reversal

Read: Figure 1 bins industries into terciles of `DownMeasure` and then plots the average intrafirm import share separately for the high-elasticity and low-elasticity groups.

- In the **complements / high-elasticity** group, the intrafirm share rises roughly from **0.39** to **0.42** to **0.47** as industries move from the first to the third tercile of downstreamness.
- In the **substitutes / low-elasticity** group, the pattern reverses: about **0.38**, **0.37**, and **0.33**.

- This is exactly the sign pattern the theory predicts.
- Why it matters: the paper is not forcing the result through a multivariate regression first. The raw data already show the sign reversal.

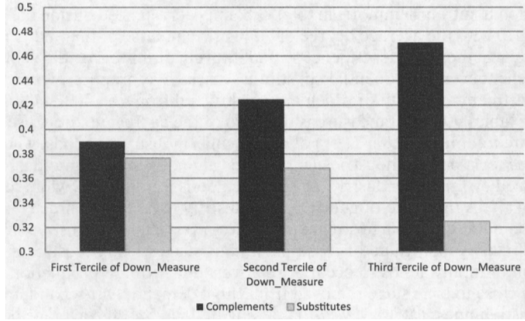


FIGURE 1.—Downstreamness and the share of intrafirm trade.

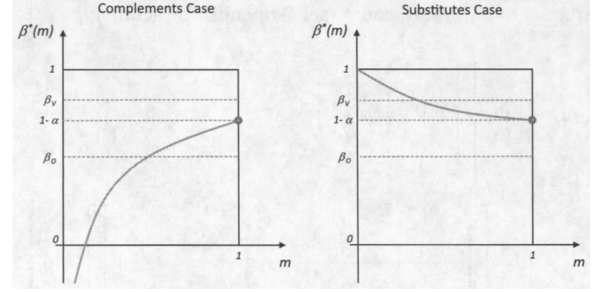


FIGURE 2.—Profit-maximizing division of surplus for stage m .

5.2 Figure 2: the optimal bargaining share along the chain

Read: Figure 2 plots the unconstrained optimal bargaining share $\beta^*(m)$ from equation (15).

- In the complements case, $\beta^*(m)$ rises with m : the firm wants more control later in the chain.
- In the substitutes case, $\beta^*(m)$ falls with m : the firm wants more control earlier in the chain.
- Why it matters: this figure is the cleanest visualization of the paper's mechanism before the discrete integrate/outsourcing choice is imposed.

5.3 Figure 3: firm heterogeneity and the integrated interval

Read: Figure 3 shows how the interval of integrated stages changes with firm productivity when integration has fixed costs.

- Low-productivity firms integrate nothing.
- As productivity rises, firms integrate more stages.
- In the complements case, the added stages are *downstream*; in the substitutes case, they are *upstream*.
- Why it matters: this is how a stage-level firm theory produces industry-level variation in related-party trade shares.

5.4 Table I: tail values of the downstreamness measures

Read: Table I lists the most and least downstream manufacturing industries according to DUse_TUse and DownMeasure.

- Broadly, primary processing industries such as fuel, chemicals, and metals look upstream, while goods closer to final use look downstream.

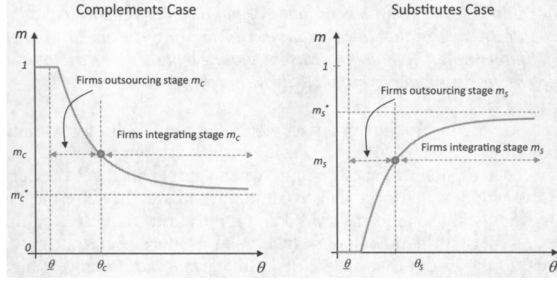


FIGURE 3.—Firm heterogeneity and the integration decision.

TABLE I
TAIL VALUES OF INDUSTRY MEASURES OF PRODUCTION LINE POSITION (DOWNSTREAMNESS)^a

IO2002	Industry	DUse_TUse IO2002	Industry	DownMeasure	
Lowest 10 values					
331314	Secondary smelting and alloying of aluminum	0.0000	325110	Petrochemical	0.2150
325110	Petrochemical	0.0599	331411	Primary smelting and refining of copper	0.2296
331411	Primary smelting and refining of copper	0.0741	331314	Secondary smelting and alloying of aluminum	0.2461
325211	Plastics material and resin	0.1205	325190	Other basic organic chemical	0.2595
325910	Printing ink	0.1325	33131A	Alumina refining and primary aluminum	0.2622
311119	Other animal food	0.1385	325310	Fertilizer	0.2658
333220	Plastics and rubber industry machinery	0.1420	335991	Carbon and graphite product	0.2668
33131A	Alumina refining and primary aluminum	0.1447	325181	Alkalies and chlorine	0.2769
335991	Carbon and graphite product	0.1615	331420	Copper rolling, drawing, extruding, and alloying	0.2769
331420	Copper rolling, drawing, extruding, and alloying	0.1804	325211	Plastics material and resin	0.2800
Highest 10 values					
334517	Irradiation apparatus	0.9669	339930	Doll, Toy, and Game	0.9705
339930	Doll, Toy, and Game	0.9686	311111	Dog and cat food	0.9717
337910	Mattress	0.9779	337910	Mattress	0.9720
322291	Sanitary paper product	0.9790	315230	Women's and girl's cut and sew apparel	0.9762
337121	Upholstered household furniture	0.9864	321991	Manufactured home (mobile home)	0.9810
337212	Office furniture and custom woodwork & millwork	0.9868	336212	Truck trailer	0.9837
336213	Motor home	0.9879	336213	Motor home	0.9879
33299A	Ammunition	0.9956	316200	Footwear	0.9927
316200	Footwear	0.9967	337121	Upholstered household furniture	0.9928
336111	Automobile	0.9997	336111	Automobile	0.9997

^aTabulated based on the set of 253 IO2002 manufacturing industries for which data on intrafirm import shares were available.

TABLE II
BASELINE DETERMINANTS OF THE INTRAFIRM IMPORT SHARE^a

	Dependent Variable: Intrafirm Import Share					
	(1)	(2)	(3)	(4)	(5)	(6)
log(<i>s</i> / <i>l</i>)	0.200*** [0.031]	0.109*** [0.038]	0.117*** [0.037]	0.143*** [0.039]	0.004 [0.043]	0.019 [0.043]
log(<i>k</i> / <i>l</i>)	0.068*** [0.014]	0.037* [0.021]		0.096*** [0.018]	0.047 [0.028]	
log(equipment <i>k</i> / <i>l</i>)			0.070*** [0.024]			0.083** [0.033]
log(plant <i>k</i> / <i>l</i>)			-0.051 [0.032]			-0.063 [0.045]
log(materials/ <i>l</i>)		0.018 [0.026]	0.023 [0.026]		0.056 [0.035]	0.063* [0.035]
log(0.001 + <i>R</i> & <i>D</i> / <i>Sales</i>)		0.029*** [0.007]	0.030*** [0.006]		0.055*** [0.009]	0.054*** [0.009]
Dispersion		0.130** [0.063]	0.150** [0.061]		0.083 [0.070]	0.120 [0.075]
Industry controls for:						
Year fixed effects?	Seller Yes	Seller Yes	Seller Yes	Buyer Yes	Buyer Yes	Buyer Yes
Observations	2783	2783	2783	2783	2783	2783
R-squared	0.23	0.30	0.31	0.17	0.27	0.28

^a ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered by industry. All columns use industry-year observations controlling for year fixed effects. Estimation is by OLS. Industry factor intensity and dispersion variables in columns 1–3 are those of the seller industry (namely, the industry that sells the input in question), while in columns 4–6, these variables are a weighted average of the characteristics of buyer industries (the industries that buy the input in question), constructed as described in Section 4.3.

- The correlation between the two measures is about **0.60**, so they agree a lot but not perfectly.
- The disagreements are informative. For example, *Fertilizer* looks fairly downstream according to DUse_TUse but much more upstream according to DownMeasure, meaning it often enters long chains despite substantial direct use.
- Why it matters: the two measures capture different aspects of position in the chain, and the results are not driven by one peculiar definition.

5.5 Table II: baseline determinants of intrafirm trade

Read: Table II establishes that the standard control variables behave sensibly and that buyer-side versions of those controls are meaningful.

- With seller controls, skill intensity is positive and significant: **0.200** in column 1, **0.109** in column 2, and **0.117** in column 3.
- Equipment capital and R&D intensity are also positive. For example, with seller controls, equipment capital is **0.070** and R&D is about **0.03**.
- With buyer controls, equipment capital and R&D remain positive and significant, while plant capital is not robustly significant.

- Dispersion is positive in several specifications, consistent with the model's prediction that more productivity dispersion raises the share of firms that integrate.
- Why it matters: before looking at downstreamness, the table shows that the rest of the empirical design lines up with earlier property-rights work and with this paper's extensions.

5.6 Table III and Figure 4: the main DUse_TUse evidence

TABLE III
DOWNSTREAMNESS AND THE INTRAFIRM IMPORT SHARE: $DUse_TUse^*$

	Dependent Variable: Intrafirm Import Share							
	(1)	(2)	(3)	(4) <i>Elas < Median</i>	(5) <i>Elas ≥ Median</i>	(6) Weighted	(7)	(8) Weighted
$\log(s/l)$	0.005 [0.044]	0.039 [0.043]	0.056 [0.042]	0.112* [0.064]	0.038 [0.055]	-0.098 [0.079]	0.005 [0.020]	-0.068 [0.076]
$\log(k/l)$	0.044 [0.029]	0.034 [0.027]						
$\log(\text{equipment } k/l)$			0.085** [0.034]	0.022 [0.047]	0.153*** [0.043]	0.188*** [0.061]	0.026 [0.016]	0.134*** [0.051]
$\log(\text{plant } k/l)$			-0.077* [0.045]	-0.011 [0.057]	-0.159** [0.064]	-0.151** [0.070]	-0.056*** [0.019]	-0.142*** [0.050]
$\log(\text{materials}/l)$	0.058* [0.035]	0.060* [0.034]	0.065* [0.033]	0.049 [0.049]	0.072 [0.047]	0.060 [0.058]	0.025* [0.014]	0.080 [0.049]
$\log(0.001 + R\&D/Sales)$	0.055*** [0.009]	0.054*** [0.009]	0.053*** [0.009]	0.050*** [0.013]	0.054*** [0.013]	0.099*** [0.018]	0.031*** [0.004]	0.073*** [0.016]
Dispersion	0.081 [0.070]	0.061 [0.070]	0.103 [0.075]	0.034 [0.108]	0.188* [0.100]	0.160 [0.124]	0.108*** [0.038]	0.083 [0.108]
$DUse_TUse$	-0.018 [0.054]			-0.216*** [0.075]	0.225*** [0.069]			
$DUse_TUse \times 1(Elas < Median), \beta_1$		-0.196*** [0.071]	-0.174** [0.072]			-0.166* [0.089]	-0.115*** [0.033]	-0.075 [0.073]
$DUse_TUse \times 1(Elas \geq Median), \beta_2$		0.171** [0.067]	0.198*** [0.068]			0.482*** [0.123]	-0.035 [0.030]	0.352*** [0.118]
$1(Elas > Median)$		-0.191*** [0.062]	-0.191*** [0.061]			-0.410*** [0.085]	-0.069* [0.029]	-0.291*** [0.075]

(Continues)

TABLE III—Continued

	Dependent Variable: Intrafirm Import Share							
	(1)	(2)	(3)	(4) <i>Elas < Median</i>	(5) <i>Elas ≥ Median</i>	(6) Weighted	(7)	(8) Weighted
p -value: Joint significance of β_1 and β_2				[0.0008]	[0.0005]		[0.0000]	[0.0021]
p -value: Test of $\beta_2 - \beta_1 = 0$				[0.0002]	[0.0001]		[0.0000]	[0.0013]
Industry controls for:	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer	Buyer
Year fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects?	No	No	No	No	No	No	No	No
Observations	2783	2783	2783	1375	1408	2783	270,991	207,991
R-squared	0.27	0.32	0.33	0.37	0.28	0.61	0.18	0.59

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered by industry. Columns 1-6 use industry-year observations controlling for year fixed effects, while columns 7-8 use country-industry-year observations controlling for country-year fixed effects. Estimation is by OLS. In all columns, the industry factor intensity and dispersion variables are a weighted average of the characteristics of buyer industries (the industries that buy the input in question), constructed as described in Section 4.3. Columns 4 and 5 restrict the sample to observations where the buyer industry elasticity is smaller (respectively, larger) than the industry median value. "Weighted" columns use the value of total imports for the industry-year or country-industry-year, respectively, as regression weights.

Read: Table III is the first core empirical test. Figure 4 plots the corresponding partial correlations after residualizing the dependent variable with respect to controls.

- Column 2, the full-sample interaction specification, gives

$$\hat{\beta}_1 = -0.196^{***}, \quad \hat{\beta}_2 = 0.177^{**}.$$

The difference is strongly significant, and the joint significance p -value is essentially zero.

- Column 3, which uses the preferred split between equipment and plant capital, yields very similar coefficients:

$$\hat{\beta}_1 = -0.174^{**}, \quad \hat{\beta}_2 = 0.198^{***}.$$

- Columns 4 and 5, which run the regressions separately by low versus high elasticity, give a direct sign reversal:

$$\text{low } Elas : -0.216^{***}, \quad \text{high } Elas : 0.225^{***}.$$

- Column 6 is the weighted baseline and is the most important quantitative result:

$$\hat{\beta}_1 = -0.166^*, \quad \hat{\beta}_2 = 0.482^{***}.$$

Using the paper's standard deviation of $DUse_TUse$ (0.228), a one-standard-deviation increase in downstreamness raises the intrafirm share in high-elasticity industries by about **0.110**.

- Column 8 shows that the country-industry-year weighted regression with country-year fixed effects still gives a strong positive complements-case slope:

$$\hat{\beta}_2 = 0.352^{***},$$

while the substitutes-case slope is small and not significant.

- Figure 4 visually mirrors the table: the residual relationship is downward sloping in the substitutes case and upward sloping in the complements case.
- Why it matters: this is the paper's headline empirical evidence.

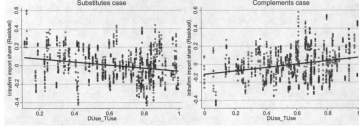


FIGURE 4.—Partial scatterplot relationship between the intrafirm trade share and $DUse_TUse$. Note: The residuals plotted on the vertical axis are predicted from an unweighted OLS regression of the intrafirm trade share on: (i) the buyer industry control variables, namely: $1(Elas > Median)$, $\log(equipm\> i)$, $\log(plant\> i)$, $\log(materials\> i)$, $\log(0.001 + R\&D/Sales)$, Dispersion, and (ii) year fixed effects. These are plotted against $DUse_TUse$ on the horizontal axis, for industry-year observations corresponding to the substitutes case ($Elas < Median$) on the left panel, and for observations from the complements case ($Elas > Median$) on the right panel.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				$Elas < Median$	$Elas > Median$	Weighted		Weighted
$\log(equipm\> i)$	-0.011 (0.005)	0.018 (0.042)	0.037 (0.031)	0.008 (0.007)	0.032 (0.031)	-0.145** (0.050)	0.002 (0.030)	-0.001* (0.016)
$\log(plant\> i)$								
$\log(materials\> i)$								
$\log(0.001 + R\&D/Sales)$								
Dispersion								
DownMeasure								
DownMeasure $\times$ $1(Elas < Median)$, β_1								
DownMeasure $\times$ $1(Elas > Median)$, β_2								

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				$Elas < Median$	$Elas > Median$	Weighted		Weighted
p-value: Joint significance of β_1 and β_2								
p-value: Test of $\beta_1 = \beta_2 = 0$								
Industry controls for								
Year fixed effects?								
Country-year fixed effects?								
Observations								
Eligibility								

5.7 Table IV: the main DownMeasure evidence

Read: Table IV repeats the core exercise using the second downstreamness measure.

- Column 2 gives a positive complements-case slope and an essentially zero substitutes-case slope:

$$\hat{\beta}_1 = 0.024, \quad \hat{\beta}_2 = 0.249^{***}.$$

- Column 3, again with the preferred control set, gives

$$\hat{\beta}_1 = 0.025, \quad \hat{\beta}_2 = 0.298^{***}.$$

- Column 5, restricting directly to high-elasticity industries, gives a strong positive slope of about **0.342*****.
- Column 6, the weighted baseline, is especially important:

$$\hat{\beta}_2 = 0.526^{***}.$$

With the standard deviation of **DownMeasure** equal to 0.222, the implied one-standard-deviation effect is about **0.117**.

- The substitutes-case slope is weaker and often imprecisely estimated here. The paper is transparent about this asymmetry: the evidence is strongest on the positive complements-case prediction.
- Why it matters: the results are not tied to one particular measure of downstreamness.

5.8 Table V: elasticity quintiles instead of a median split

Read: Table V relaxes the median split and interacts downstreamness with quintiles of the elasticity proxy.

- The positive effect of downstreamness is concentrated in the **highest elasticity quintiles**.
- The negative effect shows up in the **lowest elasticity quintile(s)**.
- The difference between top- and bottom-quintile slopes is generally significant at conventional levels.
- Why it matters: the main result is not a knife-edge artifact of choosing a median cutoff.

Downstreamness	Dependent Variable: Industry Import Share			
	(1)	(2)	(3)	(4)
	2002-2010	2002-2010	2002-2010	2002-2010
	Weighted	Weighted	Weighted	Weighted
Downstreamness × Elasticity Quintile 1, β_1	-0.027 ^{***}	-0.027 ^{***}	-0.026 ^{***}	-0.025 ^{***}
Downstreamness × Elasticity Quintile 2, β_2	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 3, β_3	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 4, β_4	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 5, β_5	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
ρ -value: Joint significance of $\beta_1, \dots, \beta_5$	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Dependent Variable: Industry Import Share											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$\beta_1 \times DOW_{it} + \beta_2 \times \text{Import Share}_{it}$	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
$\beta_1 \times DOW_{it} + \beta_2 \times \text{Import Share}_{it}$	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Value added/Value shipments	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Import "Importance"	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Intermediation	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Own contractibility	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Buyer contractibility	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
ρ -value: Joint significance of β_1 and β_2	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
ρ -value: Joint significance of β_1 and β_2	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Dependent Variable: Industry Import Share											
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Downstreamness × Elasticity Quintile 1, β_1	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}	-0.027 ^{***}
Downstreamness × Elasticity Quintile 2, β_2	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 3, β_3	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 4, β_4	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
Downstreamness × Elasticity Quintile 5, β_5	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}	-0.075 ^{***}
ρ -value: Joint significance of $\beta_1, \dots, \beta_5$	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)

5.9 Tables VI and VII: robustness to additional controls

Read: These tables add controls one by one and jointly: value-added shares, import importance, intermediation, and own/buyer contractibility.

- The core sign pattern survives throughout: downstreamness is positive in the high-elasticity group and negative or weakly negative in the low-elasticity group.
- Intermediation is strongly **negative**, consistent with the idea that wholesale intermediaries reduce the need to internalize transactions within firm boundaries.
- Inputs that are more contractible themselves tend to be traded *more* within firm boundaries, while greater buyer-industry contractibility tends to be associated with *less* integration.
- Why it matters: the downstreamness result is not proxying for these standard alternative channels.

5.10 Table VIII: headquarter-intensity extension

Read: Table VIII interacts downstreamness, the elasticity split, and quintiles of a composite headquarter-intensity measure (the first principal component of average buyer skill, equipment capital, and R&D intensity).

- The positive complements-case effect of downstreamness is strongest in the **lowest headquarter-intensity quintile**.
- There is some evidence that the negative substitutes-case effect is strongest in relatively high headquarter-intensity quintiles, though this pattern is not as sharp.
- Why it matters: this is exactly the extension suggested by Proposition 4, because higher headquarter intensity lowers effective ρ .

5.11 Table IX: firm-heterogeneity extension

Read: Table IX replaces the single downstreamness slope with downstreamness quintiles interacted with the elasticity split.

- In the high-elasticity group, the positive effect is concentrated in the **highest downstreamness quintile**.
- In the low-elasticity group, the strongest negative effect appears in the **lowest downstreamness quintile**.

TABLE VIII
EXTENSION: EFFECT OF HEADQUARTER INTENSITY*

Downstreamness:	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
HQ intensity: First principal component of buyer industry $\log(i/l)$, $\log(\text{equipment } k/l)$, and $\log(0.001 + R\&D/\text{Sales})$						
Downstream $\times 1(Elas < Med) \times (HQ \text{ Quin } 1)$, β_{11}	0.049 [0.129]	0.009 [0.139]	-0.016 [0.118]	0.351*** [0.128]	0.301** [0.134]	0.290** [0.129]
Downstream $\times 1(Elas < Med) \times (HQ \text{ Quin } 2)$, β_{12}	-0.268** [0.118]	-0.477*** [0.162]	-0.331** [0.148]	-0.161 [0.103]	-0.312* [0.186]	-0.169 [0.150]
Downstream $\times 1(Elas < Med) \times (HQ \text{ Quin } 3)$, β_{13}	-0.175 [0.148]	-0.162 [0.138]	-0.158 [0.124]	0.072 [0.131]	-0.020 [0.160]	-0.058 [0.145]
Downstream $\times 1(Elas < Med) \times (HQ \text{ Quin } 4)$, β_{14}	-0.377*** [0.134]	-0.658*** [0.108]	-0.455*** [0.069]	0.121 [0.128]	-0.394** [0.172]	-0.280** [0.125]
Downstream $\times 1(Elas < Med) \times (HQ \text{ Quin } 5)$, β_{15}	0.177 [0.171]	-0.046 [0.192]	0.024 [0.172]	0.229 [0.157]	-0.181 [0.160]	-0.050 [0.158]
Downstream $\times 1(Elas > Med) \times (HQ \text{ Quin } 1)$, β_{21}	0.196 [0.248]	0.805*** [0.325]	0.798*** [0.278]	0.465*** [0.163]	0.624*** [0.113]	0.593*** [0.087]
Downstream $\times 1(Elas > Med) \times (HQ \text{ Quin } 2)$, β_{22}	-0.037 [0.146]	-0.099 [0.181]	-0.180 [0.175]	0.032 [0.130]	-0.001 [0.238]	-0.061 [0.189]
Downstream $\times 1(Elas > Med) \times (HQ \text{ Quin } 3)$, β_{23}	0.216* [0.119]	0.172 [0.207]	0.027 [0.145]	0.558*** [0.135]	0.357 [0.257]	0.099 [0.199]
Downstream $\times 1(Elas > Med) \times (HQ \text{ Quin } 4)$, β_{24}	0.130 [0.161]	0.819*** [0.390]	0.427 [0.341]	0.280* [0.159]	0.519 [0.277]	0.079 [0.352]
Downstream $\times 1(Elas > Med) \times (HQ \text{ Quin } 5)$, β_{25}	0.183** [0.082]	0.271* [0.144]	0.162 [0.114]	0.179 [0.117]	0.274 [0.197]	0.225* [0.117]

(Continues)

- Why it matters: this fits the model with fixed costs, under which only the most downstream stages are integrated in the complements case and only the most upstream stages are integrated in the substitutes case.

TABLE IX
EXTENSION: IMPLICATIONS OF FIRM HETEROGENEITY*

Downstreamness:	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
(Downstream Quin 1) $\times 1(Elas < Median)$, β_{11}	0.101** [0.045]	0.210*** [0.058]	0.145*** [0.045]	0.060 [0.051]	0.184*** [0.059]	0.145** [0.059]
(Downstream Quin 2) $\times 1(Elas < Median)$, β_{12}	0.041 [0.043]	0.131** [0.062]	0.096* [0.054]	-0.005 [0.045]	0.062 [0.069]	0.019 [0.055]
(Downstream Quin 3) $\times 1(Elas < Median)$, β_{13}	0.008 [0.046]	0.063 [0.057]	0.010 [0.046]	-0.016 [0.049]	0.089 [0.076]	0.009 [0.060]
(Downstream Quin 4) $\times 1(Elas < Median)$, β_{14}	-0.047 [0.043]	0.092 [0.065]	0.047 [0.051]	0.024 [0.045]	0.064 [0.075]	0.032 [0.058]
(Downstream Quin 5) $\times 1(Elas < Median)$, β_{15}	0.017 [0.042]	0.084 [0.063]	0.082 [0.053]	0.023 [0.042]	0.039 [0.081]	0.058 [0.064]
(Downstream Quin 2) $\times 1(Elas > Median)$, β_{22}	0.021 [0.036]	0.031 [0.067]	0.016 [0.056]	0.013 [0.039]	-0.054 [0.069]	-0.068 [0.052]
(Downstream Quin 3) $\times 1(Elas > Median)$, β_{23}	0.086** [0.041]	0.150*** [0.065]	0.072 [0.052]	0.035 [0.047]	0.076 [0.069]	0.018 [0.066]
(Downstream Quin 4) $\times 1(Elas > Median)$, β_{24}	0.030 [0.048]	0.086 [0.056]	0.017 [0.050]	0.073* [0.043]	0.056 [0.072]	0.013 [0.060]
(Downstream Quin 5) $\times 1(Elas > Median)$, β_{25}	0.181*** [0.048]	0.352*** [0.067]	0.257*** [0.062]	0.171*** [0.059]	0.380*** [0.085]	0.300*** [0.074]

(Continues)

TABLE VIII—Continued

Downstreamness:	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
p -value: Joint significance of $\beta_{11}, \dots, \beta_{15}, \beta_{21}, \dots, \beta_{25}$	[0.0065]	[0.0000]	[0.0000]	[0.0004]	[0.0000]	[0.0000]
p -value: Test of $\beta_{21} - \beta_{25} = 0$	[0.5771]	[0.8062]	[0.8438]	[0.5397]	[0.0182]	[0.0833]
p -value: Test of $\beta_{21} - \beta_{25} = 0$	[0.9599]	[0.1790]	[0.0477]	[0.1699]	[0.1921]	[0.0262]
Additional buyer industry controls included: Main and double interaction effects, $\log(i/l)$, $\log(\text{equipment } k/l)$, $\log(\text{plant } k/l)$, $\log(\text{materials}/l)$, $\log(0.001 + R\&D/\text{Sales})$, Dispersion						
Main and double interaction effects?	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	No	No	Yes	Yes	No
Country-year fixed effects?	No	No	No	No	No	Yes
Observations	2783	2783	207,991	2783	2783	207,991
R-squared	0.40	0.69	0.63	0.40	0.71	0.65

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered by industry. Columns 1–2 and 4–5 use industry-year observations controlling for year fixed effects, while columns 3 and 6 use country-industry-year observations controlling for country-year fixed effects. Estimation is by OLS. Columns 1–3 use *DUse_TUse* and columns 4–6 use *DownMeasure* as the downstreamness variable, respectively. The log intensity measure is the first principal component of the buyer industry's $\log(i/l)$, $\log(\text{equipment } k/l)$, and $\log(0.001 + R\&D/\text{Sales})$. All columns include additional control variables whose coefficients are not reported, namely: (i) the main and double interaction effects of the buyer industry elasticity dummies and log intensity quintile dummies, and (ii) buyer industry factor intensity and dispersion variables, constructed as described in Section 4.3. "Weighted" columns use the value of total imports for the industry-year or country-industry-year, respectively, as regression weights.

TABLE IX—Continued

Downstreamness:	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
p -value: Joint significance of $\beta_{11}, \dots, \beta_{15}, \beta_{21}, \dots, \beta_{25}$	[0.0002]	[0.0000]	[0.0000]	[0.0607]	[0.0000]	[0.0000]
p -value: Test of $\beta_{21} - \beta_{25} = 0$	[0.0644]	[0.0530]	[0.1986]	[0.4766]	[0.0278]	[0.1006]
p -value: Test of $\beta_{21} - \beta_{25} = 0$	[0.0002]	[0.0000]	[0.0000]	[0.0043]	[0.0000]	[0.0001]
Additional buyer industry controls included: Main effects of Downstreamness Quintiles and Elasticity dummies, $\log(i/l)$, $\log(\text{equipment } k/l)$, $\log(\text{plant } k/l)$, $\log(\text{materials}/l)$, $\log(0.001 + R\&D/\text{Sales})$, Dispersion						
Downstream Quin and Elasticity dummies?	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	No	Yes	Yes	No
Country-year fixed effects?	No	No	Yes	No	No	Yes
Observations	2783	2783	207,991	2783	2783	207,991
R-squared	0.35	0.63	0.61	0.33	0.67	0.63

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered by industry. Columns 1–2 and 4–5 use industry-year observations controlling for year fixed effects, while columns 3 and 6 use country-industry-year observations controlling for country-year fixed effects. Estimation is by OLS. Columns 1–3 use *DUse_TUse* and columns 4–6 use *DownMeasure* as the downstreamness variable, respectively. All columns include additional control variables whose coefficients are not reported, namely: (i) the main effects of the downstreamness quintile dummies and the buyer industry elasticity dummies, and (ii) buyer industry factor intensity and dispersion variables, constructed as described in Section 4.3. "Weighted" columns use the value of total imports for the industry-year or country-industry-year, respectively, as regression weights.

5.12 Table X: selection into importing

Read: Table X applies a Heckman correction to the country-industry-year regressions.

- First stage: whether imports are positive is explained using an interaction between country entry costs and seller-industry R&D intensity.
- Entry costs significantly reduce the probability of observing positive imports.
- Second stage: the inverse Mills ratio changes the main coefficients very little.
- Why it matters: the core downstreamness results do not appear to be driven by sample selection into exporting/importing relationships.

6 Short summary

- A final-good firm sources a continuum of sequential inputs from distinct suppliers under incomplete contracts.
- The ownership decision at each stage trades off rent extraction against supplier investment incentives.

TABLE X
SELECTION INTO INTRA FIRM TRADE^a

Downstreamness	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
$\log(s/l)$	0.424** [0.166]	-0.079 [0.065]	-0.072 [0.059]	0.343** [0.167]	-0.107** [0.048]	-0.104** [0.050]
$\log(\text{equipment } k/l)$	-0.084 [0.203]	0.131*** [0.040]	0.130*** [0.041]	-0.056 [0.176]	0.125*** [0.040]	0.124*** [0.041]
$\log(\text{plant } k/l)$	-0.026 [0.234]	-0.139*** [0.046]	-0.140*** [0.046]	-0.030 [0.217]	-0.113** [0.049]	-0.113** [0.048]
$\log(\text{materials}/l)$	-0.083 [0.136]	0.063 [0.043]	0.061 [0.042]	-0.098 [0.133]	0.042 [0.042]	0.040 [0.041]
Buyer $\log(0.001 + R\&D/\text{Sales})$	-0.015 [0.064]	0.080*** [0.014]	0.080*** [0.014]	0.001 [0.061]	0.080*** [0.014]	0.080*** [0.014]
Dispersion	-0.234 [0.405]	0.117 [0.104]	0.113 [0.104]	-0.284 [0.392]	0.178 [0.115]	0.175 [0.117]
Downstream $\times 1(\text{Elas} < \text{Median})$, β_1	0.047 [0.310]	-0.073 [0.074]	-0.072 [0.074]	0.634** [0.317]	-0.020 [0.097]	-0.011 [0.088]
Downstream $\times 1(\text{Elas} > \text{Median})$, β_2	0.274 [0.252]	0.310*** [0.106]	0.314*** [0.102]	-0.291 [0.246]	0.353*** [0.078]	0.347*** [0.084]
$1(\text{Elas} > \text{Median})$	0.003 [0.231]	-0.268*** [0.075]	-0.268*** [0.075]	0.661*** [0.233]	-0.244*** [0.076]	-0.224*** [0.073]
Seller $\log(0.001 + R\&D/\text{Sales})$	-0.028*** [0.007]			-0.028*** [0.007]		
$\times$ Country Entry Costs	0.030 [0.045]			0.026 [0.041]		

(Continues)

TABLE X—Continued

Downstreamness	Dependent Variable: Intrafirm Import Share					
	(1) <i>DUse_TUse</i>	(2) <i>DUse_TUse</i> Weighted	(3) <i>DUse_TUse</i> Weighted	(4) <i>DownMeasure</i>	(5) <i>DownMeasure</i> Weighted	(6) <i>DownMeasure</i> Weighted
Inverse Mills ratio			-0.120 [0.250]		-0.082 [0.197]	
p -value: Joint significance of β_1 and β_2			[0.0036]	[0.0024]	[0.0000]	[0.0001]
p -value: Test of $\beta_2 = 0$			[0.0009]	[0.0006]	[0.0007]	[0.0008]
Industry controls for: Country-year fixed effects?	Buyer Yes	Buyer Yes	Buyer Yes	Buyer Yes	Buyer Yes	Buyer Yes
Sample:	Total imports ≥ 0	Total imports ≥ 0	Total imports ≥ 0	Total imports ≥ 0	Total imports ≥ 0	Total imports ≥ 0
Observations	462,990	186,029	186,029	462,990	186,029	186,029
R-squared	—	0.63	0.63	—	0.64	0.64

***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Standard errors are clustered by industry. Columns 1 and 4 report the first-stage probits. The exclusion restriction variable is the seller industry's R&D intensity, $\log(0.001 + R\&D/\text{Sales})$, and its interaction with a dummy variable for countries with above sample-median entry costs. The latter is constructed from the first principal component of the number of procedures, number of days, and monetary cost of starting a business, from the Doing Business data set. Columns 2 and 5 report the second stage, which is run using weighted least squares; the observation weights are the value of total imports for the country-industry-year in question. Columns 3 and 6 report the second-stage regression excluding the Inverse Mills ratio, to provide a comparison. Columns 1–3 use *DUse_TUse* and columns 4–6 use *DownMeasure* as the downstreamness variable, respectively. All columns also control for the buyer industry factor intensity and dispersion variables (constructed as described in Section 4.3), as well as country-year fixed effects.

- The new state variable is *position in the value chain*. Upstream investment affects all later stages, so the effect of integration depends on whether upstream and downstream investments are sequential complements or substitutes.
- If $\rho > \alpha$, outsource upstream and integrate downstream. If $\rho < \alpha$, integrate upstream and outsource downstream.
- The data support this sign reversal using U.S. related-party imports, IO-based downstreamness measures, and buyer-side demand elasticity proxies.
- The strongest empirical effects are positive and economically large in the high-elasticity/complements case: roughly 11–12 percentage-point increases in intrafirm shares for a one-standard-deviation rise in downstreamness.
- The paper also shows that the results survive richer bargaining assumptions, heterogeneity in firms and inputs, and a battery of empirical robustness checks.

7 Conclusion

- **Core conclusion.** The optimal boundary of the firm varies systematically along the value chain. “Make or buy” is not a single choice for the whole product; it is a stage-by-stage decision.
- **Economic intuition.** Sequential production creates propagation effects: incentives given to one supplier shape incentives for all suppliers downstream. That is why stage position matters.
- **Empirical takeaway.** Related-party trade shares are higher for downstream inputs in industries facing high buyer demand elasticities, and lower for downstream inputs in low-elasticity industries. This is a sharp and distinctive validation of the model.
- **Main limitation.** The evidence is necessarily industry-level and indirect. A cleaner test would use firm-level data that record which specific inputs are integrated and where those inputs sit in each firm’s chain.
- **Why the paper matters.** It meaningfully extends the Grossman–Hart property-rights framework from isolated transactions to a genuinely global value chain, and it gives the literature a tractable way to think about how contractual frictions interact with production sequencing.